

New Deal Policies and the Persistence of the Great Depression

A General Equilibrium Analysis

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Presentation

Introduction: The Weak Recovery

- The recovery from the Great Depression (1933–1939) was remarkably weak.
- **Data Trends:**
 - Real GDP per adult in 1939 was still 27% below trend.
 - Hours worked in 1939 were 21% below trend.
- **The Puzzle:** The large shocks (monetary, productivity, banking shocks) that are believed to have caused the crisis became positive after 1933
- **Key Question:** Why did the Great Depression persist so long?

Data: The Weak Recovery

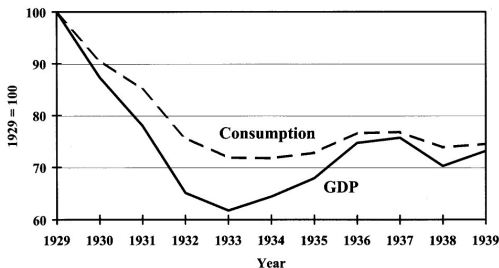


FIG. 1.—Real GDP and consumption per adult (deviations from trend)

- Real GDP and consumption per adult (deviations from trend).
- Both remain significantly depressed throughout the 1930s.

Macroeconomic Data (1934–1939)

Year	GNP	Consumption	Investment	TFP	Mfg. Wage	Private Hours
1934	64.4	71.9	27.9	92.6	111.1	68.7
1935	67.9	72.9	41.7	96.6	111.2	71.4
1936	74.7	76.7	52.6	99.9	110.5	75.8
1937	75.7	76.9	59.5	100.5	117.1	79.5
1938	70.2	73.9	38.6	100.3	122.2	71.7
1939	73.2	74.6	49.0	103.1	121.8	74.4

Table: Continuation of the Depression (1929 = 100)

- Notice that Total Factor Productivity (TFP) recovers quickly to trend (103.1 by 1939), but GNP and Hours remain low, while Manufacturing Wages soar.

Roosevelt's Economic Philosophy

"A mere builder of more industrial plants, a creator of more railroad systems, an organizer of more corporations, is as likely to be a danger as a help. Our task is not . . . necessarily producing more goods. It is the soberer, less dramatic business of administering resources and plants already in hand."

— Franklin D. Roosevelt (quoted in Kennedy 1999)

- Roosevelt believed that “excessive competition” was the root cause of the Depression.
- Strategy: Raise prices and wages by limiting competition and increasing labor bargaining power.

- **National Industrial Recovery Act (NIRA), 1933–1935:**
 - "Code of fair competition": rules for firms in an industry
 - Negotiated between the government and workers + firms
 - Permitted collusion and suspended antitrust law if firms raised wages and accepted collective bargaining.
 - NRA codes covered 80% of private, non-agricultural employment
- **National Labor Relations Act (NLRA), 1935–1939:**
 - Passed after NIRA was ruled unconstitutional.
 - Strengthened labor's bargaining power (labor union membership rose, the frequency of strikes increased).
- **Main Effect:** Sustained high real wages and high prices despite low employment.

Impact on Real Wages Across Sectors

Table: Table 2: Indexed Real Wages Relative to Trend (1929 = 100)

Sector	1930	1931	1932	1933	1934	1935	1936	1937	1938	1939
Cartelized:										
Manufacturing	101.7	106.3	105.1	102.9	110.8	112.0	111.6	118.9	122.9	123.6
Bituminous coal	101.2	104.8	91.4	90.4	110.1	119.1	125.3	127.8	130.9	132.7
Petroleum	—	—	100.0	103.6	108.9	113.6	115.4	124.8	129.1	128.8
Competitive:										
Anthracite coal	—	—	100.0	100.0	92.7	90.3	89.9	89.1	94.1	94.4
Farm	94.6	78.8	63.0	60.9	60.8	64.1	67.7	72.9	68.5	68.6

- **Significant Divergence:** After 1933, wages in cartelized sectors rose up to 33% above trend (Bituminous coal).
- **Comparison:** Anthracite coal (which failed to negotiate a code) and Farm wages remained significantly below trend.

Price Trends in Cartelized Sectors

Table: Table 5: Wholesale Prices Relative to Personal Consumption Services Deflator (Feb 1933 = 100)

Industry	Apr 1933	Dec 1933	May 1935	June 1937	June 1939
Leather/hides	102.1	131.2	127.5	128.5	121.1
Textiles	131.8	149.2	133.1	142.3	120.1
Furniture	99.4	110.3	105.3	112.2	103.0
Anthracite coal (Comp.)	91.8	91.9	80.8	78.2	77.8
Bituminous coal (Cartel)	98.4	114.1	117.0	115.6	110.1
Petroleum products	94.8	150.4	145.2	167.0	139.9
Chemicals	100.6	100.3	108.8	104.6	97.4
Iron/steel	97.9	108.2	114.6	120.2	112.6
Nonferrous metals	106.5	144.2	147.1	185.3	144.2
All metal products	99.4	107.9	109.9	115.4	110.1
Pulp/paper	98.1	114.4	108.5	122.8	101.3
All building materials	100.6	122.6	119.3	129.3	117.2
Average*	103.2	117.1	122.6	124.6	113.8

Price Trends in Cartelized Sectors

- **Abnormal Price Behavior:** Relative prices in cartelized sectors rose significantly (avg. $\sim 14\%$ by 1939).
- **Concealed Cartels:** The data supports Roosevelt's late-1930s statement that the American economy had become a "concealed cartel system... The disappearance of price competition is one of the primary causes of present difficulties"

The Model Environment: Technology

- **Intermediate Goods Producers and:**

$$y_t(i) = [z_t n_t(i)]^\gamma k_t(i)^{1-\gamma}$$

- **Sectoral Output (Y_{st}) and Final Good Production (Y_t):**

$$Y_{st} = \left[\int_{\varphi_{s-1}}^{\varphi_s} y_t(i)^\theta di \right]^{1/\theta} \quad Y_t = \left[\sum_{s=1}^S (\varphi_s - \varphi_{s-1}) Y_{st}^\phi \right]^{1/\phi}$$

- **Output of cartel sector and competitive sector:**

$$Y_{mt} = \left[\int_0^\chi y_t^\theta(i) di \right]^{1/\theta} \quad Y_{ft} = \left[\int_\chi^1 y_t^\theta(i) di \right]^{1/\theta}$$

- χ : Fraction of intermediate goods sectors that are cartelized.

The Household Problem

$$\max_{\{c_t, n_{mt}, n_{ut}, n_{ft}\}} \sum_{t=1}^{\infty} \beta^{t-1} [\log(c_t) + \psi \log(1 - n_t)]$$

Subject to:

1

$$\sum_{t=1}^{\infty} Q_t \left[w_{ft} n_{ft} + w_{mt} n_{mt} - c_t + \sum_{s=1}^S (r_{st} k_{st} - x_{st}) \right] + \Pi_1 = 0$$

2

$$n_t = n_{ft} + n_{mt} + n_{ut}$$

3

$$n_{mt} \leq \pi n_{m,t-1} + v_{t-1} n_{u,t-1}$$

4

$$k_{st+1} = x_{st} + (1 - \delta) k_{st}$$

The Cartel Mechanism: A Two-Stage Game

- We model the NIRA's link between collusion and high wages via a sequential bargaining game at the start of each period.
- **Stage 1: Insiders' Proposal.** Insiders offer a uniform wage and employment package $(\tilde{w}_t, \tilde{n}_t)$.
- **Stage 2: Firms' Response.**
 - *Accept:* Firms operate as a legal cartel, constrained to hire $\tilde{n}_t$ at $\tilde{w}_t$.
 - *Reject:* Firms hire from the competitive spot market at w_{ft} . Crucially, the government enforces antitrust (preventing monopoly pricing) with probability $1 - \omega$.
- **The Policy Friction:** Monopoly pricing is the “carrot” used to enforce above-market union wages, distorting labor allocation.

The Firm's Profit and Participation Constraint

- **Monopoly Profit Function:**

$$\Pi_t(w) = \max_{n,k} \{ Y_t^{1-\phi} Y_{mt}^{\phi-\theta} [(z_t n)^\gamma k^{1-\gamma}]^\theta - r_t k - w n \}$$

- **Reservation Profit:**

$$P_t = \omega \Pi_t(w_{ft})$$

- **The Role of ω :** The parameter $\omega \in [0, 1]$ indexes the firm's relative bargaining power. A low ω reflects a high probability of antitrust enforcement if firms reject the union, giving labor significant leverage.

The Insiders' Dynamic Problem

Insiders maximize the expected present value of the wage premium, choosing $(\tilde{w}, \tilde{n})$ subject to the firm's participation constraint:

$$V(n) = \max_{(\tilde{w}, \tilde{n})} \left\{ \min \left[1, \frac{\tilde{n}}{n} \right] \left[\tilde{w}_t - w_{ft} + \pi \frac{Q_{t+1}}{Q_t} V_{t+1}(\pi \tilde{n}) \right] \right\}$$

Subject to:

$$\Pi_t(\tilde{w}, \tilde{n}) \geq P_t$$

Proposition 1: The optimal policy is such that:

- (i) $\Pi_t(\tilde{w}, \tilde{n}) = P_t$
- (ii) if $n \leq n_t^*$, then $\tilde{n} \geq n$
- (iii) if $n_t^* < n \leq N_t(w_{ft})$, then $\tilde{n} = n$
- (iv) if $n > N_t(w_{ft})$, then $\tilde{n} \leq n$

Steady-State Implications: The Role of Bargaining Power

Table: Table 6: Cartel Model Steady-State Variables Relative to Competitive Model

		Output	Employment	Cartel Wage	Cartel Employment	Fraction of Searchers
$\chi = 0.25$						
$\omega = 1.00$	$\phi = 0$	0.97	0.98	0.96	0.91	0.00
	$\phi = -1$	0.97	0.98	0.96	0.94	0.00
	$\phi = -2$	0.97	0.98	0.96	0.96	0.00
$\omega = 0.50$	$\phi = 0$	0.94	0.96	1.04	0.82	0.01
	$\phi = -1$	0.94	0.95	1.04	0.87	0.01
	$\phi = -2$	0.95	0.95	1.04	0.89	0.01
$\omega = 0.05$	$\phi = 0$	0.86	0.90	1.35	0.57	0.04
	$\phi = -1$	0.85	0.88	1.34	0.67	0.05
	$\phi = -2$	0.86	0.87	1.34	0.70	0.06
$\chi = 0.50$						
$\omega = 0.05$	$\phi = 0$	0.76	0.81	1.18	0.58	0.09
	$\phi = -1$	0.75	0.79	1.16	0.65	0.11
	$\phi = -2$	0.75	0.78	1.15	0.67	0.11

Quantitative Evaluation: Output Recovery

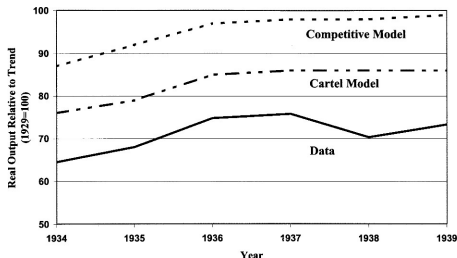


FIG. 2.—Output in the data and in the models

- **Neoclassical Benchmark:** Predicts a rapid return to trend driven by rebounding TFP.
- **Cartel Model:** The insider-outsider friction severely dampens the recovery, closely mirroring the empirical stagnation.

Transitional Dynamics: Competitive vs. Cartel

Table: Table 7: Equilibrium Path from the Competitive Model

Year	Output	Consumption	Investment	Employment	Wage
1934	0.87	0.90	0.73	0.98	0.89
1935	0.92	0.91	0.97	1.01	0.91
1936	0.97	0.93	1.18	1.03	0.94
1937	0.98	0.94	1.14	1.03	0.95
1938	0.98	0.95	1.12	1.02	0.96
1939	0.99	0.96	1.09	1.02	0.97

Table: Table 8: Simulated Transition Path for Cartel Model ($\chi = 0.32, \omega = 0.10$)

Year	Output	Consumption	Investment	Employment	Cartel Wage	Comp. Wage
1934	0.77	0.85	0.40	0.82	1.16	0.81
1935	0.81	0.85	0.62	0.84	1.19	0.83
1936	0.86	0.85	0.87	0.89	1.20	0.83
1937	0.87	0.86	0.90	0.90	1.20	0.83
1938	0.86	0.86	0.86	0.89	1.20	0.84
1939	0.87	0.86	0.88	0.89	1.20	0.84

- **Mechanism:** New Deal policies (NIRA/NLRA) explicitly linked industry collusion to above-market wages, creating a severe insider-outsider friction.
- **Quantitative Contribution:** This cartelization mechanism accounts for approximately **60%** of the output shortfall during the 1933–1939 recovery.
- **Historical Perspective:** The standard neoclassical framework fails to explain the 1930s stagnation. Incorporating the explicit political economy distortions of the New Deal allows us to map the precise architecture of the weak recovery.